

UNSUPERVISED MACHINE LEARNING

120 Final Exam Practice Questions & Answer Key

Topic 1: Overview & Fundamentals of Unsupervised Learning

Q1: How does unsupervised learning fundamentally differ from supervised learning?

- A. It uses labeled data for validation instead of training.
- B. It operates on predictor features without any associated supervising output label y .
- C. It focuses strictly on predicting continuous target variables.
- D. It requires twice as many training observations to converge.

Correct Answer: B — It operates on predictor features without any associated supervising output label y .

Q2: Why is model evaluation inherently more subjective in unsupervised learning?

- A. Supervised algorithms are non-deterministic, while unsupervised ones are strictly deterministic.
- B. Unsupervised models do not have true output labels y against which predictions can be validated.
- C. Unsupervised learning cannot be implemented in Python or R.
- D. Cross-validation is mathematically impossible for any matrix operation.

Correct Answer: B — Unsupervised models do not have true output labels y against which predictions can be validated.

Q3: Which of the following is a primary task of unsupervised learning?

- A. Classifying emails as spam or not spam based on labeled ground truth.
- B. Predicting next month's stock market returns using historical response values.
- C. Discovering latent subgroups or clusters among unlabeled customer records.
- D. Estimating logistic regression coefficients for binary medical diagnoses.

Correct Answer: C — Discovering latent subgroups or clusters among unlabeled customer records.

Q4: What type of statistical problem is Semi-Supervised Learning?

- A. A setting where all predictor features X are missing.
- B. A setting where a small subset m of n observations have response labels y , while $n-m$ lack labels.
- C. A method that uses reinforcement learning to label unsupervised clusters.
- D. An algorithm that splits data into equal training and test sets without features.

Correct Answer: B — A setting where a small subset m of n observations have response labels y , while $n-m$ lack labels.

Q5: Which metric CANNOT be directly computed to evaluate unsupervised clustering quality?

- A. Within-cluster sum of squares (WCSS / Inertia).
- B. Average Silhouette score.
- C. Classification Test Accuracy against ground truth.
- D. Average inter-cluster dissimilarity distance.

Correct Answer: C — Classification Test Accuracy against ground truth.

Q6: In market segmentation, why is clustering preferred over supervised regression when customer spending labels are unavailable?

- A. Clustering automatically creates new numerical response variables for each customer.
- B. Clustering partitions customers into homogenous groups based on observed feature similarities without requiring pre-existing spending labels.
- C. Clustering guarantees finding the single global optimum for revenue prediction.
- D. Supervised regression cannot process more than two predictor variables.

Correct Answer: B — Clustering partitions customers into homogenous groups based on observed feature similarities without requiring pre-existing spending labels.

Q7: What is meant by model 'stability' in unsupervised learning validation?

- A. The model produces identical R-squared scores across different regression fits.
- B. The discovered patterns or cluster structures remain consistent across random seeds, data subsamples, or slight perturbations.
- C. The algorithm executes in linear $O(N)$ computational time regardless of dataset size.
- D. The model eliminates all noise from raw high-dimensional features.

Correct Answer: B — The discovered patterns or cluster structures remain consistent across random seeds, data subsamples, or slight perturbations.

Q8: Which of the following scenarios represents an unsupervised learning application?

- A. Forecasting sales volume based on advertising expenditures in TV, Radio, and Newspaper.
- B. Grouping 64 cancer cell lines based on 6,830 gene expression measurements to find cell line subtypes.
- C. Predicting whether a bank customer will default on a credit card loan.
- D. Estimating a patient's risk of heart disease based on clinical trial measurements.

Correct Answer: B — Grouping 64 cancer cell lines based on 6,830 gene expression measurements to find cell line subtypes.

Q9: How can unsupervised learning complement a downstream supervised learning pipeline?

- A. By fabricating ground truth response labels for unobserved test samples.
- B. By performing dimensionality reduction (e.g. PCA) to reduce overfitting and speed up classifier training.
- C. By replacing cross-validation with deterministic matrix inversions.
- D. By converting all quantitative predictors into qualitative dummy variables.

Correct Answer: B — By performing dimensionality reduction (e.g. PCA) to reduce overfitting and speed up classifier training.

Q10: What is the main danger of using highly flexible non-parametric methods on small unsupervised datasets?

- A. High variance and sensitivity to random noise or sample perturbations.
- B. Underfitting due to forced linear decision boundaries.
- C. Inability to compute pairwise Euclidean distances.
- D. Excessive memory consumption during matrix transposition.

Correct Answer: A — High variance and sensitivity to random noise or sample perturbations.

Q11: In exploratory data analysis, what is the 'curse of dimensionality'?

- A. High-dimensional space causes all pairwise data point distances to become nearly equal and sparse.
- B. Having more observations than predictors causes matrix inversion failures.
- C. Decision tree algorithms cannot process continuous numeric features.
- D. Variance explained by PCA drops to zero in 2 dimensions.

Correct Answer: A — High-dimensional space causes all pairwise data point distances to become nearly equal and sparse.

Q12: Why must domain expertise be combined with mathematical metrics when evaluating unsupervised outputs?

- A. Unsupervised algorithms lack semantic domain awareness and cannot automatically assign meaningful business labels to statistical subgroups.
- B. Mathematical metrics are mathematically invalid for datasets with more than 3 features.
- C. Domain experts can force algorithms to converge to zero training error.
- D. Internal metrics like WCSS always increase when data is standardized.

Correct Answer: A — Unsupervised algorithms lack semantic domain awareness and cannot automatically assign meaningful business labels to statistical subgroups.

Topic 2: Distance Metrics & Feature Standardization

Q13: What happens if you perform Euclidean distance-based clustering on unstandardized data where Feature A ranges from 0 to 1 and Feature B ranges from 0 to 1,000?

- A. Feature A will completely dominate the distance calculations.
- B. Feature B will completely dominate the distance calculations due to its large numerical magnitude.
- C. Both features will contribute equally to the distance matrix.
- D. The Euclidean distance calculation will throw a runtime linear algebra error.

Correct Answer: B — Feature B will completely dominate the distance calculations due to its large numerical magnitude.

Q14: What is the mathematical formula for Z-score standardization of a feature X_j ?

- A. $z_j = (x_j - \min(x_j)) / (\max(x_j) - \min(x_j))$
- B. $z_j = (x_j - \text{mean}(x_j)) / \text{std}(x_j)$
- C. $z_j = \log(x_j) - \text{mean}(x_j)$
- D. $z_j = x_j^2 / \text{var}(x_j)$

Correct Answer: B — $z_j = (x_j - \text{mean}(x_j)) / \text{std}(x_j)$

Q15: Which distance metric is most robust to extreme outliers because it does not square coordinate differences?

- A. Euclidean distance (L2 norm)
- B. Manhattan distance (L1 norm)
- C. Squared Mahalanobis distance
- D. Frobenius matrix norm

Correct Answer: B — Manhattan distance (L1 norm)

Q16: When comparing text document term frequency vectors or sparse recommendation ratings, which distance measure is most appropriate?

- A. Euclidean distance
- B. Cosine distance
- C. Manhattan distance
- D. Single linkage distance

Correct Answer: B — Cosine distance

Q17: What does correlation-based distance focus on when comparing two observation vectors x_i and x_j ?

- A. The absolute Euclidean distance in raw units.
- B. The shape and relative profile of feature values across dimensions rather than their absolute magnitudes.
- C. The number of missing entries in the dataset.
- D. The ratio of sample size N to number of predictors P .

Correct Answer: B — The shape and relative profile of feature values across dimensions rather than their absolute magnitudes.

Q18: In the USArrests dataset (Murder, Assault, UrbanPop, Rape), why does unstandardized PCA/clustering focus almost entirely on Assault?

- A. Assault has the highest mean value.
- B. Assault has by far the largest raw variance (6945.16) among all four features.
- C. Assault is a categorical qualitative variable.
- D. Assault is perfectly uncorrelated with Murder and Rape.

Correct Answer: B — Assault has by far the largest raw variance (6945.16) among all four features.

Q19: Which transformation is most effective for skewed features with heavy right tails (e.g. household income or expenditure)?

- A. Min-Max normalization
- B. Logarithmic transformation $y = \log(x)$
- C. Squaring feature values $y = x^2$
- D. Reversing feature sign $y = -x$

Correct Answer: B — Logarithmic transformation $y = \log(x)$

Q20: How does Gower's distance handle datasets containing both continuous and categorical features?

- A. It converts continuous features into binary indicator variables before computing Manhattan distance.
- B. It computes a weighted average of normalized metric dissimilarities for continuous variables and non-matching indicators for categorical variables.
- C. It drops all categorical variables and performs standard Euclidean distance on continuous variables.
- D. It applies principal component analysis directly to qualitative text strings.

Correct Answer: B — It computes a weighted average of normalized metric dissimilarities for continuous variables and non-matching indicators for categorical variables.

Q21: Which distance metric is specifically used to compare binary or categorical vectors of equal length by counting mismatches?

- A. Minkowski distance
- B. Hamming distance
- C. Euclidean distance
- D. Mahalanobis distance

Correct Answer: B — Hamming distance

Q22: What is the mean and standard deviation of any feature after applying Z-score standardization?

- A. Mean = 1.0, Standard Deviation = 0.0
- B. Mean = 0.0, Standard Deviation = 1.0
- C. Mean = 0.5, Standard Deviation = 0.5
- D. Mean = Min Value, Standard Deviation = Max Value

Correct Answer: B — Mean = 0.0, Standard Deviation = 1.0

Q23: What is the relationship between Manhattan distance and Euclidean distance between two distinct points x and y ?

- A. Manhattan distance is always strictly less than Euclidean distance.
- B. Manhattan distance is always greater than or equal to Euclidean distance ($L1 \geq L2$).
- C. Manhattan distance equals zero whenever Euclidean distance is positive.
- D. They are mathematically identical for all multi-dimensional vectors.

Correct Answer: B — Manhattan distance is always greater than or equal to Euclidean distance ($L1 \geq L2$).

Q24: Why does collinearity (two highly correlated features) distort distance-based clustering?

- A. It effectively double-counts the underlying shared attribute, giving it twice the weight in distance calculations.
- B. It forces all distance values to become negative.
- C. It causes the covariance matrix to become identity.
- D. It removes all outliers from the dataset.

Correct Answer: A — It effectively double-counts the underlying shared attribute, giving it twice the weight in distance calculations.

Topic 3: PCA — Foundations & Mathematical Formulation

Q25: What is the primary objective when deriving the first principal component loading vector ϕ_1 ?

- A. To minimize the total variance of the projected score values.
- B. To maximize the sample variance of the linear combination Z_1 subject to $\sum(\phi_{j1}^2) = 1$.
- C. To ensure all loading values are strictly positive.
- D. To force the principal component to pass through the maximum value of X .

Correct Answer: B — To maximize the sample variance of the linear combination Z_1 subject to $\sum(\phi_{j1}^2) = 1$.

Q26: Why is data centering (column means = 0) a mandatory prerequisite before applying PCA?

- A. Because linear models require non-negative predictors.
- B. Because the principal component vector originates at the origin and has no intercept term.
- C. To force all singular values to equal 1.0.
- D. To convert quantitative variables into categorical dummy variables.

Correct Answer: B — Because the principal component vector originates at the origin and has no intercept term.

Q27: What constraint is imposed on the principal component loading vector $\phi_1 = (\phi_{11}, \dots, \phi_{p1})^T$?

- A. $\sum(\phi_{j1}) = 0$
- B. $\sum(\phi_{j1}^2) = 1$ (unit length constraint)
- C. $\phi_{j1} \geq 0$ for all j
- D. $\prod(\phi_{j1}) = 1$

Correct Answer: B — $\sum(\phi_{j1}^2) = 1$ (unit length constraint)

Q28: What are the 'scores' z_{i1} in Principal Component Analysis?

- A. The p-values associated with feature significance tests.
- B. The projections (coordinates) of observation i onto the first principal component loading vector direction.
- C. The proportion of missing values in row i .
- D. The classification error rates of a KNN model.

Correct Answer: B — The projections (coordinates) of observation i onto the first principal component loading vector direction.

Q29: What geometric property holds between the first principal component Z_1 and the second principal component Z_2 ?

- A. They are parallel lines in feature space.
- B. They are orthogonal (perpendicular / uncorrelated) to each other.
- C. They intersect at a 45-degree angle.
- D. They have equal loading values for all features.

Correct Answer: B — They are orthogonal (perpendicular / uncorrelated) to each other.

Q30: Besides maximizing sample variance, what is the dual geometric interpretation of PCA?

- A. Finding the linear hyperplane that minimizes the sum of squared perpendicular Euclidean distances from data points to the hyperplane.
- B. Finding the bounding box that encompasses all data points.
- C. Minimizing the total number of observations N .
- D. Maximizing the correlation between independent features.

Correct Answer: A — Finding the linear hyperplane that minimizes the sum of squared perpendicular Euclidean distances from data points to the hyperplane.

Q31: How is SVD (Singular Value Decomposition) $X = U * \text{Sigma} * V^T$ used to compute PCA on a centered data matrix X ?

- A. The left singular vectors U contain the loadings, and V contains the residuals.
- B. The right singular vectors V (columns) contain the principal component loading vectors, and $U * \text{Sigma}$ yields the component score vectors.
- C. Sigma contains the misclassification error rates.
- D. U converts continuous features into categorical classes.

Correct Answer: B — The right singular vectors V (columns) contain the principal component loading vectors, and $U * \text{Sigma}$ yields the component score vectors.

Q32: What is the maximum possible number of distinct principal components for an $N \times P$ matrix X ?

- A. $N + P$
- B. $\min(N - 1, P)$
- C. $N * P$
- D. Always exactly 2

Correct Answer: B — $\min(N - 1, P)$

Q33: What is the covariance matrix of standardized data matrix X ?

- A. The identity matrix I .
- B. The correlation matrix R of the original dataset.
- C. A diagonal matrix containing only zeroes.
- D. The inverse of the loading matrix V^{-1} .

Correct Answer: B — The correlation matrix R of the original dataset.

Q34: In Singular Value Decomposition $X = U * \text{Sigma} * V^T$, what are the mathematical properties of matrices U and V ?

- A. They are diagonal matrices.
- B. They are orthonormal matrices ($U^T * U = I$ and $V^T * V = I$).
- C. They contain only positive integer values.
- D. They represent classification confusion matrices.

Correct Answer: B — They are orthonormal matrices ($U^T * U = I$ and $V^T * V = I$).

Q35: What does the j -th loading ϕ_{j1} tell us about the relationship between feature X_j and component Z_1 ?

- A. The exact mean value of X_j .
- B. The weight, contribution, and (when standardized) correlation of feature X_j with principal component Z_1 .
- C. The number of missing entries in column X_j .
- D. The standard deviation of component Z_1 .

Correct Answer: B — The weight, contribution, and (when standardized) correlation of feature X_j with principal component Z_1 .

Q36: Why is PCA considered an unsupervised technique?

- A. Because it requires a binary response label Y to compute singular values.
- B. Because it operates solely on the predictor matrix X without relying on any outcome or supervising response variable Y .
- C. Because it can only be evaluated using cross-validation test error.
- D. Because loading vectors are computed using decision trees.

Correct Answer: B — Because it operates solely on the predictor matrix X without relying on any outcome or supervising response variable Y .

Topic 4: PCA — Variance Explained, Scree Plots & Biplots

Q37: What does a biplot display simultaneously in a single 2D visual layout?

- A. Training MSE and Test MSE curves across iterations.
- B. Principal component scores (observations as points) and loading vectors (features as directional arrows).
- C. Cluster dendrograms and silhouette width bars.
- D. Eigenvalues and residual error rates.

Correct Answer: B — Principal component scores (observations as points) and loading vectors (features as directional arrows).

Q38: What is a 'scree plot' used for in Principal Component Analysis?

- A. To visualize the decision boundary of a support vector machine.
- B. To plot the Proportion of Variance Explained (PVE) for each component to identify an 'elbow' for dimensionality selection.
- C. To plot missing values across matrix rows.
- D. To calculate pairwise Manhattan distances.

Correct Answer: B — To plot the Proportion of Variance Explained (PVE) for each component to identify an 'elbow' for dimensionality selection.

Q39: What is the mathematical definition of Proportion of Variance Explained (PVE) for the m-th component?

- A. $PVE_m = \sum(z_m^2) / \sum_j(\sum_i(x_{ij}^2))$
- B. $PVE_m = \text{Mean}(Z_m) / \text{Var}(Z_m)$
- C. $PVE_m = \sum(\phi_{jm})$
- D. $PVE_m = 1 - \text{MSE}$

Correct Answer: A — $PVE_m = \sum(z_m^2) / \sum_j(\sum_i(x_{ij}^2))$

Q40: What is the sum of PVEs across all $M = \min(N-1, P)$ principal components?

- A. 0.50
- B. 1.00 (100% of total variance)
- C. P
- D. N

Correct Answer: B — 1.00 (100% of total variance)

Q41: Are principal component loading vectors unique across different software packages?

- A. Yes, they are strictly unique down to exact floating point signs.
- B. They are unique up to a sign flip (ϕ_m and $-\phi_m$ represent the same line/axis direction).
- C. No, every software package produces completely different orthogonal directions.
- D. Loadings change every time the random seed is updated.

Correct Answer: B — They are unique up to a sign flip (ϕ_m and $-\phi_m$ represent the same line/axis direction).

Q42: What happens to the reconstructed matrix product $z_m * \phi_{jm}$ if the signs of both score vector Z_m and loading vector ϕ_m are flipped?

- A. The product flips sign and corrupts the data.
- B. The product remains completely unchanged because $(-1) * (-1) = 1$.
- C. The total variance explained drops by 50%.
- D. The matrix becomes non-invertible.

Correct Answer: B — The product remains completely unchanged because $(-1) * (-1) = 1$.

Q43: In the USArrests biplot, what does it mean if Murder, Assault, and Rape arrows point in the same direction along PC1?

- A. They are negatively correlated with each other.
- B. They are positively correlated with each other and jointly contribute to high positive PC1 scores (overall crime rate).
- C. UrbanPop has the largest variance among all features.
- D. PC1 captures only noise.

Correct Answer: B — They are positively correlated with each other and jointly contribute to high positive PC1 scores (overall crime rate).

Q44: When using PCA as a preprocessing step for Principal Components Regression (PCR), how should M (number of components) be chosen objectively?

- A. By selecting M where training MSE is zero.
- B. By treating M as a tuning parameter and evaluating test MSE via Cross-Validation.
- C. By keeping all P components.
- D. By choosing M based on plot colors.

Correct Answer: B — By treating M as a tuning parameter and evaluating test MSE via Cross-Validation.

Q45: What does a flat cumulative PVE curve (e.g. PC1 explains 10%, PC2 explains 8%, etc.) suggest about the dataset?

- A. The features are highly correlated and lie on a low-dimensional line.
- B. The dataset lacks strong linear structure, and many components are required to capture the variance.
- C. Data standardization was performed correctly.
- D. The dataset contains zero missing values.

Correct Answer: B — The dataset lacks strong linear structure, and many components are required to capture the variance.

Q46: What is the relationship between PCA PVE and the R^2 of low-rank matrix approximation?

- A. PVE equals $1 - (RSS / TSS)$, representing the R^2 of approximating X with M components.
- B. PVE is completely unrelated to RSS or TSS.
- C. PVE measures $1 - \text{Accuracy}$.
- D. PVE equals the F-statistic of multiple regression.

Correct Answer: A — PVE equals $1 - (RSS / TSS)$, representing the R^2 of approximating X with M components.

Q47: In the Wine dataset PCA, if PC1 has high positive loadings for Flavanoids and Phenols, and negative loading for Malic Acid, how is PC1 interpreted?

- A. A contrast between high quality/flavor profile (positive) versus high acidity (negative).
- B. PC1 measures only bottle volume.
- C. Flavanoids and Malic Acid are perfectly collinear.
- D. Malic Acid dominates total dataset variance.

Correct Answer: A — A contrast between high quality/flavor profile (positive) versus high acidity (negative).

Q48: Why is eyeballing an 'elbow' on a scree plot considered an ad-hoc method?

- A. Because scree plots cannot be generated for matrices with more than 10 rows.
- B. Because there is no universal, mathematically objective definition of where an 'elbow' occurs in unsupervised exploratory settings.
- C. Because PVE values are non-numeric.
- D. Because PCA requires supervised target labels to draw scree plots.

Correct Answer: B — Because there is no universal, mathematically objective definition of where an 'elbow' occurs in unsupervised exploratory settings.

Topic 5: Non-Linear Dimensionality Reduction & Manifold Learning

Q49: Why does linear PCA fail to preserve the true structure of non-linear manifolds like a 'Swiss Roll' dataset?

- A. PCA cannot handle datasets with more than 2 features.
- B. PCA assumes global linear structure and measures straight-line Euclidean distances through empty space rather than distances along the curved manifold.
- C. PCA requires binary response variables.
- D. PCA cannot be computed via SVD.

Correct Answer: B — PCA assumes global linear structure and measures straight-line Euclidean distances through empty space rather than distances along the curved manifold.

Q50: What type of distances does Isomap compute to capture the true non-linear structure along a manifold?

- A. Mahalanobis distances
- B. Geodesic distances (shortest path along a neighborhood graph constructed over the manifold)
- C. Hamming distances
- D. Chebyshev distances

Correct Answer: B — Geodesic distances (shortest path along a neighborhood graph constructed over the manifold)

Q51: How does t-SNE (t-Distributed Stochastic Neighbor Embedding) address the 'crowding problem' in low-dimensional visualizations?

- A. By replacing Gaussian distributions in high dimensions with a Student-t distribution (heavy tails) in low dimensions.
- B. By forcing all data points into a single circle.
- C. By setting all pairwise distances to zero.
- D. By applying linear regression to scores.

Correct Answer: A — By replacing Gaussian distributions in high dimensions with a Student-t distribution (heavy tails) in low dimensions.

Q52: What objective function does t-SNE minimize to align low-dimensional similarity distribution Q with high-dimensional similarity P?

- A. Mean Squared Error (MSE)
- B. Kullback-Leibler (KL) Divergence
- C. Cross-Entropy Loss
- D. Residual Sum of Squares (RSS)

Correct Answer: B — Kullback-Leibler (KL) Divergence

Q53: Is t-SNE a deterministic algorithm?

- A. Yes, running t-SNE multiple times always yields identical coordinates.
- B. No, t-SNE is non-deterministic because it relies on random initialization and stochastic gradient descent optimization.
- C. Yes, because it uses exact SVD matrix decomposition.
- D. Yes, if the number of features $P > 100$.

Correct Answer: B — No, t-SNE is non-deterministic because it relies on random initialization and stochastic gradient descent optimization.

Q54: What is the primary input required to perform Classical Multidimensional Scaling (MDS)?

- A. A matrix of predictor features X and outcome labels Y.
- B. A pairwise dissimilarity/distance matrix D between all observations.
- C. A set of decision tree split rules.
- D. A trained neural network model.

Correct Answer: B — A pairwise dissimilarity/distance matrix D between all observations.

Q55: What mathematical step is performed in Classical MDS to convert squared distances D^2 into Gram matrix B of inner products?

- A. Z-score standardization
- B. Double-centering using centering matrix $H = I - (1/n) \cdot \mathbf{1} \mathbf{1}^T$
- C. Logarithmic transformation
- D. Min-Max scaling

Correct Answer: B — Double-centering using centering matrix $H = I - (1/n) \cdot \mathbf{1} \mathbf{1}^T$

Q56: What is a major limitation of t-SNE regarding distance interpretation across distant clusters?

- A. Distances between distant clusters in a t-SNE plot are not meaningful or reliable (only local neighborhoods are preserved).
- B. Local cluster neighborhoods are completely destroyed.
- C. It cannot run on datasets with more than 10 rows.
- D. It requires supervised classification targets.

Correct Answer: A — Distances between distant clusters in a t-SNE plot are not meaningful or reliable (only local neighborhoods are preserved).

Q57: What advantage does Isomap have over t-SNE regarding algorithm reproducibility?

- A. Isomap is deterministic (the same graph and shortest path computations yield identical embeddings every time).
- B. Isomap executes in $O(1)$ time.
- C. Isomap does not require computing distances.
- D. Isomap eliminates the need for standardization.

Correct Answer: A — Isomap is deterministic (the same graph and shortest path computations yield identical embeddings every time).

Q58: In MDS, if two points are twice as far apart on the MDS plot as two other points, how should that distance be interpreted?

- A. The points are approximately twice as dissimilar in original high-dimensional dissimilarity space.
- B. The points belong to the same decision tree leaf.
- C. The points have zero variance.
- D. The points are false positives.

Correct Answer: A — The points are approximately twice as dissimilar in original high-dimensional dissimilarity space.

Q59: What neighborhood construction methods are used in Step 1 of Isomap?

- A. K-Nearest Neighbors (k-NN) graph or epsilon-ball neighborhood graph
- B. Decision tree entropy splits
- C. Linear regression residuals
- D. K-means centroid assignment

Correct Answer: A — K-Nearest Neighbors (k-NN) graph or epsilon-ball neighborhood graph

Q60: What is the computational complexity of computing all-pairs shortest paths for Isomap on N data points?

- A. $O(N)$
- B. $O(N^2 \log N)$ or $O(N^3)$ (e.g., Floyd-Warshall / Dijkstra algorithms)
- C. $O(\log N)$
- D. $O(1)$

Correct Answer: B — $O(N^2 \log N)$ or $O(N^3)$ (e.g., Floyd-Warshall / Dijkstra algorithms)

Topic 6: K-Means Clustering — Algorithm & Theory

Q61: What is the primary mathematical objective of the K-means clustering algorithm?

- A. To maximize the inter-cluster distance between all centroids.
- B. To minimize the Within-Cluster Sum of Squares (WCSS / Inertia) across all K clusters.
- C. To maximize the total number of clusters K .
- D. To minimize the total correlation among features.

Correct Answer: B — To minimize the Within-Cluster Sum of Squares (WCSS / Inertia) across all K clusters.

Q62: Why is the coordinate-wise mean the mathematically optimal choice for updating a cluster centroid in K-means?

- A. Because the mean maximizes the variance within the cluster.
- B. Because the sample mean is the unique constant that minimizes the sum of squared Euclidean distances to all points in that cluster.
- C. Because the mean eliminates feature correlations.
- D. Because the mean prevents cluster overlap.

Correct Answer: B — Because the sample mean is the unique constant that minimizes the sum of squared Euclidean distances to all points in that cluster.

Q63: What guarantee does the K-means algorithm provide regarding its convergence?

- A. It is guaranteed to find the global minimum of the objective function J .
- B. It is guaranteed to converge to a local minimum (or plateau) in a finite number of steps because each step strictly decreases or maintains J .
- C. It guarantees that all clusters will contain an equal number of observations.
- D. It guarantees 100% classification accuracy.

Correct Answer: B — It is guaranteed to converge to a local minimum (or plateau) in a finite number of steps because each step strictly decreases or maintains J .

Q64: What happens during the 'Assignment Step' of the K-means iteration?

- A. Centroids are re-estimated by taking column means.
- B. Each observation x_i is assigned to the cluster C_k whose centroid μ_k is closest in Euclidean distance.
- C. Random noise is added to feature columns.
- D. The number of clusters K is increased by 1.

Correct Answer: B — Each observation x_i is assigned to the cluster C_k whose centroid μ_k is closest in Euclidean distance.

Q65: What happens during the 'Update Step' of the K-means iteration?

- A. Data points are randomly re-assigned to new clusters.
- B. Each centroid μ_k is recalculated as the coordinate-wise mean of all observations currently assigned to cluster C_k .
- C. Features are re-standardized using Z-scores.
- D. Outliers are removed from the matrix.

Correct Answer: B — Each centroid μ_k is recalculated as the coordinate-wise mean of all observations currently assigned to cluster C_k .

Q66: Why is running K-means with multiple random initializations ($n_init = 20$ or 50) strongly recommended?

- A. Because K-means can converge to different local minimum solutions depending on initial centroid placement.
- B. Because K increases automatically during each initialization.
- C. To convert Euclidean distances into Manhattan distances.
- D. Because single runs always throw runtime exception errors.

Correct Answer: A — Because K-means can converge to different local minimum solutions depending on initial centroid placement.

Q67: What shape of clusters does the K-means objective function inherently favor?

- A. Elongated, non-convex spiral shapes.
- B. Roughly spherical (isotropic) clusters of equal variance and similar size.
- C. Concentric ring shapes.
- D. Hierarchical tree structures.

Correct Answer: B — Roughly spherical (isotropic) clusters of equal variance and similar size.

Q68: How are two clusters properties defined in K-means partitioning $C_1, \dots, C_K$?

- A. Non-overlapping ($C_k \cap C_{k'} = \text{empty}$) and exhaustive ($\cup C_k = \{1, \dots, n\}$).
- B. Overlapping soft probability densities.
- C. Nested sub-branches of a dendrogram tree.
- D. Independent random variables.

Correct Answer: A — Non-overlapping ($C_k \cap C_{k'} = \text{empty}$) and exhaustive ($\cup C_k = \{1, \dots, n\}$).

Q69: What is 'Inertia' in scikit-learn's KMeans implementation?

- A. The number of iterations required to converge.
- B. The total Within-Cluster Sum of Squares (WCSS).
- C. The ratio of N to P .
- D. The silhouette score of PC1.

Correct Answer: B — The total Within-Cluster Sum of Squares (WCSS).

Q70: How does K-means++ improve upon standard random initialization?

- A. It selects initial centroids probabilistically proportional to their squared distance from existing centroids, spreading them out in feature space.
- B. It sets all initial centroids to the origin (0,0).
- C. It uses decision trees to pre-classify observations.
- D. It eliminates the update step completely.

Correct Answer: A — It selects initial centroids probabilistically proportional to their squared distance from existing centroids, spreading them out in feature space.

Q71: Why is K-means sensitive to extreme outliers?

- A. Because squaring Euclidean distances causes extreme outliers to heavily pull centroids away from true cluster centers.
- B. Because outliers force K to equal 1.
- C. Because outliers convert quantitative variables into strings.
- D. Because outliers prevent matrix multiplication.

Correct Answer: A — Because squaring Euclidean distances causes extreme outliers to heavily pull centroids away from true cluster centers.

Q72: What happens to WCSS (Inertia) as K increases from 1 to N?

- A. WCSS increases monotonically.
- B. WCSS decreases monotonically, reaching exactly 0 when $K = N$ (each point is its own centroid).
- C. WCSS remains constant.
- D. WCSS oscillates unpredictably.

Correct Answer: B — WCSS decreases monotonically, reaching exactly 0 when $K = N$ (each point is its own centroid).

Topic 7: Cluster Evaluation, Profiling & Determining K
Q73: What does the Silhouette Score compute for a clustered data point i ?

- A. The ratio of total variance explained by PC1.
- B. A comparison between mean intra-cluster distance $a(i)$ and mean nearest-cluster separation distance $b(i)$: $s(i) = (b(i) - a(i)) / \max(a(i), b(i))$.
- C. The p-value of a two-sample t-test between cluster means.
- D. The perpendicular distance to the maximal margin hyperplane.

Correct Answer: B — A comparison between mean intra-cluster distance $a(i)$ and mean nearest-cluster separation distance $b(i)$: $s(i) = (b(i) - a(i)) / \max(a(i), b(i))$.

Q74: When profiling clusters of human populations or customer data, what is a key ethical consideration?

- A. Maximizing the total number of clusters.
- B. Ensuring labels do not perpetuate social stereotypes, stigmatize subgroups, or introduce personal bias.
- C. Using raw unstandardized features for labeling.
- D. Ignoring cluster sizes completely.

Correct Answer: B — Ensuring labels do not perpetuate social stereotypes, stigmatize subgroups, or introduce personal bias.

Q75: What is the range of the Silhouette Coefficient $s(i)$?

- A. $[0, \infty)$
- B. $[-1, +1]$
- C. $[0, 100]$
- D. $(-\infty, 0]$

Correct Answer: B — $[-1, +1]$

Q76: What does a Silhouette score close to +1 indicate for an observation i ?

- A. The observation is misclustered and lies closer to a neighboring cluster.
- B. The observation is highly cohesive within its own cluster and well-separated from neighboring clusters.
- C. The observation lies exactly on the boundary between two clusters.
- D. The observation contains missing values.

Correct Answer: B — The observation is highly cohesive within its own cluster and well-separated from neighboring clusters.

Q77: What does a negative Silhouette score $s(i) < 0$ indicate?

- A. The point is well-clustered.
- B. The point has been assigned to the wrong cluster (closer to neighbor cluster on average than its assigned cluster).
- C. K is optimal.
- D. Feature standardization succeeded.

Correct Answer: B — The point has been assigned to the wrong cluster (closer to neighbor cluster on average than its assigned cluster).

Q78: In the Elbow Method for determining K , what are you looking for on the plot of WCSS vs K ?

- A. The highest peak on the curve.
- B. The point of diminishing returns ('elbow') where WCSS reduction flattens significantly.
- C. The point where WCSS equals 100.
- D. The point where $K = N$.

Correct Answer: B — The point of diminishing returns ('elbow') where WCSS reduction flattens significantly.

Q79: How is 'cluster profiling' performed after fitting a clustering algorithm?

- A. By deleting all clusters with fewer than 10 points.
- B. By calculating feature summary statistics (means, medians, standard deviations) per cluster to understand feature distributions across groups.
- C. By predicting future stock prices.
- D. By setting all cluster labels to random integers.

Correct Answer: B — By calculating feature summary statistics (means, medians, standard deviations) per cluster to understand feature distributions across groups.

Q80: What is the Gap Statistic used for in cluster evaluation?

- A. Comparing total within-cluster variation for different K against their expected values under a null reference distribution of unclustered data.
- B. Filling in missing entries in rating matrices.
- C. Measuring the execution time of SVD algorithms.
- D. Calculating kernel loss in Support Vector Machines.

Correct Answer: A — Comparing total within-cluster variation for different K against their expected values under a null reference distribution of unclustered data.

Q81: When profiling clusters, why should summary statistics be inspected on ORIGINAL scales rather than Z-standardized scales?

- A. Standardized z-scores are mathematically invalid for means.
- B. Original feature units (e.g. \$95,000 income, 42 years age) are domain-interpretable for human decision makers.
- C. Standardized scales change cluster assignments.
- D. Z-scores hide negative numbers.

Correct Answer: B — Original feature units (e.g. \$95,000 income, 42 years age) are domain-interpretable for human decision makers.

Q82: What visual tool projects high-dimensional cluster assignments onto a 2D scatter plot for inspection?

- A. Histogram of raw values
- B. 2D PCA score scatter plot colored by cluster assignment labels
- C. Confusion matrix heatmap
- D. ROC curve display

Correct Answer: B — 2D PCA score scatter plot colored by cluster assignment labels

Q83: Why is choosing K based strictly on minimizing WCSS problematic without an elbow or gap analysis?

- A. Because WCSS continuously decreases as K increases, reaching 0 when $K = N$.
- B. Because WCSS cannot be calculated for $K > 2$.
- C. Because WCSS is non-numeric.
- D. Because WCSS requires supervised test labels.

Correct Answer: A — Because WCSS continuously decreases as K increases, reaching 0 when $K = N$.

Q84: How can silhouette plots help identify the quality of individual clusters?

- A. By displaying silhouette widths for every sample grouped by cluster, revealing if specific clusters contain misclustered or wide/narrow distributions.
- B. By displaying the R^2 of linear regression.
- C. By showing execution times across CPU cores.
- D. By counting missing entries per column.

Correct Answer: A — By displaying silhouette widths for every sample grouped by cluster, revealing if specific clusters contain misclustered or wide/narrow distributions.

Topic 8: Hierarchical Clustering & Linkage Criteria

Q85: How should similarity between two observations be interpreted on a dendrogram?

- A. By looking at horizontal proximity along the bottom leaf axis.
- B. By looking at the vertical height at which branches containing the two observations first fuse.
- C. By counting the total number of leaves in the tree.
- D. By measuring the thickness of the vertical line branches.

Correct Answer: B — By looking at the vertical height at which branches containing the two observations first fuse.

Q86: Which linkage criterion defines dissimilarity as the MAXIMUM pairwise distance between an observation in Cluster A and an observation in Cluster B?

- A. Single linkage
- B. Complete linkage
- C. Average linkage
- D. Centroid linkage

Correct Answer: B — Complete linkage

Q87: What structural issue is Single Linkage prone to, where clusters grow by repeatedly attaching single nearby observations?

- A. Inversion
- B. Chaining
- C. Crowding
- D. Overfitting

Correct Answer: B — Chaining

Q88: How does Agglomerative Hierarchical Clustering build clusters?

- A. Top-down, starting with one cluster containing all points and splitting recursively.
- B. Bottom-up, starting with N singleton clusters and iteratively merging the least dissimilar pair.
- C. By calculating decision tree classification rules.
- D. By estimating Gaussian mixture proportions.

Correct Answer: B — Bottom-up, starting with N singleton clusters and iteratively merging the least dissimilar pair.

Q89: Which linkage criterion computes the AVERAGE pairwise distance between all observations in Cluster A and Cluster B?

- A. Single linkage
- B. Complete linkage
- C. Average linkage
- D. Centroid linkage

Correct Answer: C — Average linkage

Q90: What unique drawback occurs in Centroid Linkage hierarchical clustering?

- A. It requires K to be specified prior to running.
- B. Inversions can occur, where two clusters fuse at a vertical height BELOW one or both of their daughter nodes.
- C. It cannot be used with Euclidean distance.
- D. It destroys all leaf node orderings.

Correct Answer: B — Inversions can occur, where two clusters fuse at a vertical height BELOW one or both of their daughter nodes.

Q91: How do you extract a partition of K clusters from a completed hierarchical dendrogram?

- A. By deleting the bottom leaves of the tree.
- B. By making a horizontal cut across the dendrogram at a specific vertical height threshold.
- C. By re-running standard K-means on the leaf labels.
- D. By multiplying the dissimilarity matrix by its transpose.

Correct Answer: B — By making a horizontal cut across the dendrogram at a specific vertical height threshold.

Q92: Why does horizontal proximity between leaves on a dendrogram NOT imply similarity?

- A. Because leaf ordering along the horizontal axis can be flipped ($2^{(N-1)}$ possible reorderings) at any merge node without changing tree topology or distance meanings.
- B. Because horizontal distances represent time in seconds.
- C. Because dendrograms are plotted in log scale.
- D. Because leaves are placed in alphabetical order.

Correct Answer: A — Because leaf ordering along the horizontal axis can be flipped ($2^{(N-1)}$ possible reorderings) at any merge node without changing tree topology or distance meanings.

Q93: What is the computational complexity of standard agglomerative hierarchical clustering on N observations?

- A. $O(N)$
- B. $O(N^2)$ or $O(N^3)$ due to pairwise dissimilarity matrix updates
- C. $O(\log N)$
- D. $O(1)$

Correct Answer: B — $O(N^2)$ or $O(N^3)$ due to pairwise dissimilarity matrix updates

Q94: Why might Hierarchical Clustering yield LESS accurate results than K-means if the true data groups are NOT nested?

- A. Hierarchical clustering enforces a strict nested structure on cuts, whereas K-means partitions data flexibly without nesting constraints.
- B. Hierarchical clustering cannot handle continuous features.
- C. Hierarchical clustering requires zero missing values.
- D. Hierarchical clustering always sets $K = 1$.

Correct Answer: A — Hierarchical clustering enforces a strict nested structure on cuts, whereas K-means partitions data flexibly without nesting constraints.

Q95: Which two linkage criteria tend to produce the most balanced, compact dendrograms in statistical practice?

- A. Single linkage and Centroid linkage
- B. Complete linkage and Average linkage
- C. Random linkage and Uniform linkage
- D. Min-Max linkage and Median linkage

Correct Answer: B — Complete linkage and Average linkage

Q96: In Genomics gene expression analysis, why is Correlation-Based Distance preferred over Euclidean Distance for hierarchical clustering?

- A. Because gene profiles with similar relative expression shapes (up-regulation / down-regulation across conditions) cluster together regardless of baseline expression magnitude.
- B. Because Euclidean distance cannot process numeric arrays.
- C. Because genomic data has zero standard deviation.
- D. Because correlation-based distance runs in $O(1)$ time.

Correct Answer: A — Because gene profiles with similar relative expression shapes (up-regulation / down-regulation across conditions) cluster together regardless of baseline expression magnitude.

Topic 9: Missing Data Imputation & Matrix Completion

Q97: In Algorithm 12.1 (Iterative SVD Matrix Completion), what happens to the OBSERVED entries in the matrix X_{tilde} during each iteration?

- A. They are overwritten with SVD predictions.
- B. They remain fixed at their original observed values.
- C. They are set to NaN.
- D. They are converted to Z-scores.

Correct Answer: B — They remain fixed at their original observed values.

Q98: In a recommender system rating matrix (Users x Movies), how are the M components in $x_{\text{hat}}_{ij} = \sum_m (a_{\text{hat}}_{im} * b_{\text{hat}}_{jm})$ interpreted?

- A. a_{hat}_{im} is user membership/preference in clique/genre m ; b_{hat}_{jm} is movie strength in genre m .
- B. a_{hat}_{im} is movie length; b_{hat}_{jm} is user age.
- C. a_{hat}_{im} is classification test error; b_{hat}_{jm} is training MSE.
- D. a_{hat}_{im} is Euclidean distance; b_{hat}_{jm} is Manhattan distance.

Correct Answer: A — a_{hat}_{im} is user membership/preference in clique/genre m ; b_{hat}_{jm} is movie strength in genre m .

Q99: What crucial assumption about missing data pattern is required for low-rank SVD matrix completion to yield unbiased estimates?

- A. Missing Not at Random (MNAR)
- B. Missing at Random (MAR) — missingness does not depend on the unobserved missing values themselves.
- C. Every row must be 100% missing.
- D. Features must be completely uncorrelated.

Correct Answer: B — Missing at Random (MAR) — missingness does not depend on the unobserved missing values themselves.

Q100: What is the initial baseline imputation used in Step 1 of Algorithm 12.1 before running iterative SVD?

- A. Zero imputation
- B. Column mean imputation (filling missing x_{ij} with non-missing mean of feature j)
- C. Random integer assignment
- D. Deleting all incomplete rows

Correct Answer: B — Column mean imputation (filling missing x_{ij} with non-missing mean of feature j)

Q101: How is the optimal rank M chosen in iterative matrix completion?

- A. By selecting $M = P$.
- B. By masking a subset of observed entries as a validation set, running completion across candidate ranks M , and selecting rank M with lowest validation RMSE.
- C. By picking $M = 0$.
- D. By setting M equal to dataset row count N .

Correct Answer: B — By masking a subset of observed entries as a validation set, running completion across candidate ranks M , and selecting rank M with lowest validation RMSE.

Q102: What is the convergence criterion in Algorithm 12.1 for matrix completion?

- A. When K-means inertia reaches zero.
- B. When the relative change in objective function $\sum((x_{ij} - \sum(a_{\hat{im}} * b_{\hat{jm}}))^2)$ falls below a small tolerance threshold (e.g. $1e-7$).
- C. When all values become integers.
- D. When 100 iterations pass regardless of error.

Correct Answer: B — When the relative change in objective function $\sum((x_{ij} - \sum(a_{\hat{im}} * b_{\hat{jm}}))^2)$ falls below a small tolerance threshold (e.g. $1e-7$).

Q103: In the Netflix dataset example, what percentage of the user-movie rating matrix was typically missing?

- A. 10%
- B. Around 99% sparse (most customers rate only a tiny fraction of the total movie catalog).
- C. 0%
- D. Exactly 50%

Correct Answer: B — Around 99% sparse (most customers rate only a tiny fraction of the total movie catalog).

Q104: Why does low-rank SVD matrix completion outperform simple column mean imputation?

- A. Low-rank SVD exploits pairwise correlations across features and rows to infer missing entries, whereas column mean imputation ignores feature correlations.
- B. Column mean imputation takes $O(N^3)$ computational time.
- C. Low-rank SVD deletes incomplete rows.
- D. SVD requires supervised labels.

Correct Answer: A — Low-rank SVD exploits pairwise correlations across features and rows to infer missing entries, whereas column mean imputation ignores feature correlations.

Q105: Before applying matrix completion to a user-movie rating matrix, why is column/row centering performed?

- A. To convert ratings into binary indicators.
- B. To subtract baseline user rating tendencies or movie popularity, allowing low-rank SVD to focus on preference deviations.
- C. To force all ratings to equal zero.
- D. To eliminate matrix dimensions.

Correct Answer: B — To subtract baseline user rating tendencies or movie popularity, allowing low-rank SVD to focus on preference deviations.

Q106: How is imputation quality validated in experimental settings when ground truth is known?

- A. By computing Pearson correlation r and Root Mean Squared Error (RMSE) between true held-out ratings and imputed values.
- B. By counting standard deviations.
- C. By measuring execution time in seconds.
- D. By plotting ROC curves.

Correct Answer: A — By computing Pearson correlation r and Root Mean Squared Error (RMSE) between true held-out ratings and imputed values.

Q107: What is 'Hard-Impute' in matrix completion literature?

- A. An iterative algorithm that truncates singular values to rank M at each step to update missing entries.
- B. Deleting all rows with missing entries.
- C. Filling missing entries with extreme outliers.
- D. A supervised random forest classifier.

Correct Answer: A — An iterative algorithm that truncates singular values to rank M at each step to update missing entries.

Q108: Can standard PCA functions (e.g. `sklearn.decomposition.PCA`) be applied directly to a raw matrix containing NaN values?

- A. Yes, scikit-learn PCA ignores NaN natively.
- B. No, standard PCA algorithms require a complete matrix without missing entries, necessitating matrix completion or imputation first.
- C. Yes, if $P < 5$.
- D. Yes, if data is scaled.

Correct Answer: B — No, standard PCA algorithms require a complete matrix without missing entries, necessitating matrix completion or imputation first.

Detection

Q109: What fundamental principle do Isolation Forests exploit for anomaly detection?

- A. Anomalies are close to cluster centroids.
- B. Anomalies lie in sparse regions and require FEWER random decision tree splits (shorter path lengths) to isolate than normal points.
- C. Anomalies always have zero variance.
- D. Anomalies require deep trees with over 100 splits.

Correct Answer: B — Anomalies lie in sparse regions and require FEWER random decision tree splits (shorter path lengths) to isolate than normal points.

Q110: In an Isolation Forest, if the contamination parameter is set to 0.05 (5%), but only 1% of points are truly anomalous, what will happen?

- A. The model will flag exactly 1% of points.
- B. The model will flag 5% of points, labeling some normal observations as anomalies (false positives).
- C. The model will fail to compile.
- D. Path lengths will become negative.

Correct Answer: B — The model will flag 5% of points, labeling some normal observations as anomalies (false positives).

Q111: What parameters define each Gaussian component k in a Gaussian Mixture Model (GMM)?

- A. Mean vector μ_k , Covariance matrix Σ_k , and Mixing weight/prior π_k .
- B. Decision tree depth and leaf count.
- C. Learning rate and momentum.
- D. Slope and intercept.

Correct Answer: A — Mean vector μ_k , Covariance matrix Σ_k , and Mixing weight/prior π_k .

Q112: What takes place during the 'Expectation Step' (E-step) of the EM algorithm for GMMs?

- A. Model parameters μ_k , Σ_k are updated to maximize likelihood.
- B. Posterior probabilities ('responsibilities') $w_{ik} = P(C_k | x_i)$ are calculated for each observation given current parameter estimates.
- C. Data points are permanently removed from memory.
- D. K is incremented by 1.

Correct Answer: B — Posterior probabilities ('responsibilities') $w_{ik} = P(C_k | x_i)$ are calculated for each observation given current parameter estimates.

Q113: What takes place during the 'Maximization Step' (M-step) of the EM algorithm for GMMs?

- A. Responsibilities w_{ik} are computed.
- B. Model parameters μ_k , Σ_k , π_k are updated using weighted sample statistics to maximize expected complete-data log-likelihood.
- C. Initial random seeds are re-selected.
- D. Decision boundaries are linearized.

Correct Answer: B — Model parameters μ_k , Σ_k , π_k are updated using weighted sample statistics to maximize expected complete-data log-likelihood.

Q114: How does a Gaussian Mixture Model (GMM) differ from K-means regarding cluster assignments?

- A. GMM provides soft, probabilistic assignments $P(C_k | x_i)$, whereas K-means performs hard, deterministic assignments.
- B. GMM requires binary features, whereas K-means requires strings.
- C. GMM cannot fit elliptical shapes.
- D. K-means uses likelihood maximization.

Correct Answer: A — GMM provides soft, probabilistic assignments $P(C_k | x_i)$, whereas K-means performs hard, deterministic assignments.

Q115: In text preprocessing for Latent Dirichlet Allocation (LDA), what is 'tokenization'?

- A. Converting document text strings into individual word tokens.
- B. Removing rare document rows.
- C. Squaring term frequencies.
- D. Fitting linear regression lines.

Correct Answer: A — Converting document text strings into individual word tokens.

Q116: How does Latent Dirichlet Allocation (LDA) represent documents and topics?

- A. Documents are represented as decision tree splits.
- B. Documents are represented as mixtures of topics (θ), and topics are represented as probability distributions over vocabulary words (ϕ).
- C. Documents are represented as single integer IDs.
- D. Topics are deterministic binary labels.

Correct Answer: B — Documents are represented as mixtures of topics (θ), and topics are represented as probability distributions over vocabulary words (ϕ).

Q117: How are image pixels ($8 \times 8 = 64$ pixels) formatted as input for a Generative GMM on digit images?

- A. They are flattened into a 64-dimensional numerical feature vector per image.
- B. They are converted into text strings.
- C. They are plotted as a 1D line chart.
- D. They are passed directly into standard linear regression without flattening.

Correct Answer: A — They are flattened into a 64-dimensional numerical feature vector per image.

Q118: How does a trained GMM generate brand NEW synthetic digit image samples?

- A. By copying and pasting existing training images.
- B. By sampling a mixture component k according to weights π , and then drawing a random vector $X \sim N(\mu_k, \Sigma_k)$.
- C. By setting all pixel values to 255.
- D. By running logistic regression on pixel indices.

Correct Answer: B — By sampling a mixture component k according to weights π , and then drawing a random vector $X \sim N(\mu_k, \Sigma_k)$.

Q119: In an Isolation Forest, what path length $h(x)$ is expected for a normal, non-anomalous observation inside a dense cluster?

- A. A very short path length $h(x)$ (1 or 2 splits).
- B. A deep path length $h(x)$ requiring many random recursive splits to isolate from surrounding neighbors.
- C. A negative path length.
- D. An infinite path length.

Correct Answer: B — A deep path length $h(x)$ requiring many random recursive splits to isolate from surrounding neighbors.

Q120: What normalized anomaly score $S(x)$ threshold in an Isolation Forest indicates a high likelihood of being an anomaly?

- A. $S(x)$ close to 0.0
- B. $S(x)$ close to 0.5 (average point)
- C. $S(x)$ close to 1.0 (significantly shorter path length than expected)
- D. $S(x) < -1.0$

Correct Answer: C — $S(x)$ close to 1.0 (significantly shorter path length than expected)

